

Monitoring Cloud Instances with Ganglia and sFlow

This article looks into some quick steps for installing, configuring and working with Ganglia and sFlow to monitor your cloud instances.



Today's modern IT environment comprises a vast pool of heterogeneous infrastructure resources that are either on-premise or in the cloud. Monitoring such complex environments can be a real challenge owing to the fast-paced, ever increasing complexity of resources.

A typical cloud-based data centre can have thousands of physical servers spread across multiple geographic locations, each running a hypervisor with one or more virtual machines on it. Each of these virtual machines further connects to a virtual or physical switch, storage arrays, etc. Real-time performance monitoring and visibility becomes quite essential in such diverse and complex environments.

This article looks into the combined use of Ganglia and sFlow as a monitoring solution for a dynamic cloud environment.

Introducing Ganglia

Ganglia is an open source monitoring tool that was initially designed to monitor high performance computing systems such as grids and clusters. It is highly scalable by design and allows IT administrators to get a holistic view of the IT environment's performance.

Ganglia leverages multicast-based protocols to listen to and advertise the state of the machines within the cluster it is monitoring. It uses XML for data representation, the XDR (External Data Representation) format for transporting

the metric data, and an open source storage and visualisation tool called *RRDtool*. A combination of these makes Ganglia a truly concurrent and robust monitoring tool.

The Ganglia monitoring system collects and processes metric data using two daemons or services, namely *gmond* and *gmetad*.

Gmond (Ganglia Monitoring Daemon)

Gmond is a simple daemon that runs on every host that has to be monitored within a cluster. It is designed to have very little overhead on the host it is monitoring. It is very easy to install and configure, supporting both Linux as well as Windows operating systems (as per its latest release v3.5.7). *Gmond* primarily monitors and announces the state change of the host to the *gmetad* daemon using XML over unicast or multicast channels.

Gmetad (Ganglia Metadata Daemon)

This daemon is primarily responsible for polling *gmond* daemons across a specific cluster, gathering the XML data, parsing it and saving the data in the round-robin database (RRD). This data can then be advertised to a client over a TCP socket.

The *gmetad* daemon is designed to collect metric data from multiple *gmond* as well as other *gmetad* daemons. This type of scenario is best suited for cloud environments in which there are multiple clusters of servers spread across geographical regions. Each cluster can have at least one *gmetad* daemon that polls the